Heart Disease Predictions using CDC Survey Data

Danie le Roux 2489056

Supervisor(s):
Dr. Dambala Gelo Kutela

A Capstone Project report submitted in partial fulfillment of the requirements for the degree of Master of Science in the field of e-Science

in the

School of Computer Science and Applied Mathematics
University of the Witwatersrand, Johannesburg

30 November 2022

Declaration

I, Danie le Roux 2489056, declare that this report is my own, unaided work. It is being submitted for the degree of Master of Science in the field of e-Science at the University of the Witwatersrand, Johannesburg. It has not been submitted for any degree or examination at any other university.

Danie le Roux 2489056

30 November 2022

Abstract

Heart disease is a very serious ailment with many factors that lead to its development. The primary goal of this research project is to investigate the factors that contribute to heart disease and to develop a machine-learning model that can aid in its prediction. The model could assist medical professionals to predict the likelihood of a patient developing heart disease by looking at contributing risk factors and common co-morbidities. This research project makes use of various machine-learning approaches to identify significant factors and to create a model with the highest prediction accuracy. 2022 Survey data collected by the Centres for Disease Control and Prevention (CDC), using their Behavioral Risk Factor Surveillance System (BRFSS), was used to construct the models. The research project focused on F2 scores, Recall, to determine the model with the best fit. Ultimately, a Logistic Regression model with reduced dimensionality using Principal Component Analyses (PCA) was selected, the model could very accurately predict True Negatives but due to an imbalanced class problem, only 5,6 percent of sample entries had heart disease, the model could not accurately predict True Positives. While the model often predicted False Positives, this was seen as beneficial since it could be used as an indicator to medical professionals that the patient exhibited many high-risk factors that that contribute to developing heart disease and can thus be used as an early pre-emptive indicator that the patient has high risk of developing the condition.

Acknowledgements

I would like to thank Dr. Kutela for the opportunity to work on this study and for his guidance and assistance. I would also like to thank Dr. Robertson for her assistance and understanding. Lastly, I would like to thank the NEPTTP for this great opportunity to complete my masters.

Contents

List of Figures

List of Tables

Chapter 1

Introduction

According to the Centers for Disease Control and Prevention (CDC) (2022), heart disease is a serious condition that affects the blood flow to the heart and can lead to Arrhythmia, heart failure and heart attacks due to a decrease in blood flow. The factors identified by the CDC that greatly contribute to heart disease consist of diabetes, obesity, physical inactivity, excessive alcohol use, and an unhealthy diet.

1.1 Background

1.1.1 Using Machine Learning to Analyse Medical Conditions

According to Kumar et al. (2022), machine learning greatly assists medical practitioners with intelligent health systems and patient care. They found that machine learning significantly improved the infirmary experience and is essential in the healthcare industry, particularly for illness detection and analyses. The study focused on how machine learning techniques improved illness diagnosis areas and were beneficial for the diagnosis and prevention of cancer, heart disease, stroke, hypertension, skin disease, Alzheimer's, chronic disease, and cerebrovascular disease.

A study conducted by Javaid et al. (2022) found that machine learning applications have a significant impact on the healthcare system by improving the accuracy and speed of medical professionals. Machine learning applications can aid in the detection of early risk indicators for a variety of medical conditions. Furthermore, such algorithms allow medical professionals to focus on patient care rather than capturing and searching for important information required to make a diagnosis, greatly improving the efficiency of the healthcare system while simultaneously lowering the cost of care.

1.1.2 Application of Machine Learning in Healthcare

Advancements in machine learning and artificial intelligence revolutionised identification and prediction in healthcare, effectively improving patient treatment. Machine learning has further led to an increase in the precision, quality, and speed that medical professionals can diagnose and treat patients (Habehh and Gohel, 2021).

1.1.3 Heart Disease

Adams et al. (2018) conducted a study where they examined the benefits of early detection and diagnosis of heart disease. The study found that early detection of heart disease significantly improved a patient's prognosis and assisted medical professionals in aiding patients with the condition. It is thus crucial to develop methods for the early treatment and detection of heart disease (Duprez, 2006).

1.2 Problem Statement

The problem that this research aims to identify is what the most significant factors that contribute to the development of heart disease are and how a model that can accurately identify patients with a high risk of developing the condition can be constructed. Identifying the most significant risk factors that lead to the development of heart disease will allow the model to accurately predict high-risk patients. This model can be used by medical professionals to predict the development of heart disease before it takes place and will thus allow medical practitioners to take preventive steps in assisting the patient.

1.3 Research Question

The primary question that this research project aims to answer is whether it is possible to construct a machine learning model that makes use of identified significant high-risk factors to accurately predict heart disease in patients before the ailment develops, allowing medical professionals to take preventive steps to assist the patient before the ailment becomes too serious.

1.4 Research Aims and Objectives

1.4.1 Research Aims

This research project aims to develop a model that can identify the likelihood of a patient developing heart disease. This will be done by first identifying the most significant contributing factors and then constructing a machine learning model to use these factors to predict the likelihood of a patient developing the ailment. This could greatly assist medical professionals in identifying patients that are likely to develop the condition due to present co-morbidities.

1.4.2 Objectives

The objectives of this research project are:

- To construct a data set containing medical information of patients from CDC survey data.
- To identify factors that are significantly related to the development of heart disease.
- To construct a model that is capable of predicting the likelihood of a patient developing heart disease.

1.5 Limitations

The research project will have the following limitations:

- Due to the small proportion of the population that develops heart disease, there will be an imbalanced classification problem that will need to be accounted for by making use of under-sampling.
- The CDC survey data was collected from American patients, this could cause a lack of generalisability to patients outside of America, data will need to be collected from a wider population to improve generalisability.

1.6 Assumptions and Definitions

The following important definitions are relevant to this research project:

- Co-morbidity: The presence of two or more simultaneous medical conditions in a patient.
- Precision: Ratio of true positives (TP) against all positive predictions.

$$P = TP/(TP + FP) \tag{1.1}$$

- Recall: Ratio of true positives to all positive cases.

$$R = ((1 + \beta^2) \times P \times R)/(\beta^2 \times P + R) \tag{1.2}$$

- F2 Score: Model performance score that is weighted to recall.
- Principal Component Analyses (PCA): A dimensionality-reduction method that transforms a large set of variables into a smaller one without losing most of the information.
- Logistic Regression: A model that looks at the probability of an event taking place based on the log odds of the event.
- K-Nearest Neighbours Algorithm (KNN): A model that makes use of the k nearest training examples to determine the probability of an event taking place.
- Random Forest: A model that constructs and averages multiple decision trees during training that determines the probability of an event taking place through the average likelihoods of all the decision trees.
- Naive Bayes Classifier: A model that applies Bayes′ Theorem to create probabilistic classifiers used in determining the likelihood of an event taking place.
- Multi-Layer Perceptron Classifier (MLP): A neural network classifier that makes use of an underlying neural network to classify outcomes.
- Easy Ensemble Classifier: A classifier that consists of an ensemble of AdaBoost learners that are trained on balanced bootstrap samples.

1.7 Overview

This research project made use of survey data collected by the CDC to identify high-risk factors that contributed to the development of heart disease. A variety of models were constructed and compared using F2 Scores, focusing on recall, to obtain a model with the best accuracy in predicting a patient's likelihood of developing heart disease. Ultimately the Logistic Regression model was selected due to best performance (True Positive predictions were better compared to a random sampler – 59 percent vs 50 percent). The Logistic Regression model very accurately predicted True Negatives but had a high likelihood of predicting False Positives, this can, however, be used as an indicator for medical professionals that the patient displays many symptoms that could lead to heart disease and was thus found to be acceptable.

Chapter 2

Research Methodology

2.1 Research design

An exploratory experimental method was used in conducting this research project. Significant features related to heart disease were explored and identified and then used to construct a model that can predict the likelihood of a patient having heart disease. Due to there being no prior assumptions as to what features would be significant, the research was exploratory in nature.

2.2 Data

2022 Survey data collected on patients by the CDC was used in identifying significant factors that lead to heart disease. The dataset contained a large number of dimensions in the feature space (278 features excluding labels) and consisted of 401 958 samples where only 5,6 percent of entries had heart disease. The feature space consisted of highly relevant features such as a patient's general health to less relevant features such as whether the patient owned a cell phone or a landline. Initially, an intuition-based reduction was used to remove all irrelevant features (e.g. cell phone vs. landline). A BMI feature was recalculated from the provided patient weight and height information.

Due to responses in the survey being voluntary, a large number of samples had null entries. Samples with null entries were either deleted or imputed with feature medians where appropriate. Sample outliers were identified and imputed with feature means, see Table 2.1.

Treatments:

- A - Delete samples containing null entries
- B - Impute null, "don't know", and "refuse" entries with feature medians
- C - Impute outliers with feature means
- D - Dropping features
- E - One Hot Encoding

TABLE 2.1: Null Treatment

Feature	Number of Nulls	Treatment
HeartDisease	3	A
BMI	41357	D + Recalculated
GeneralHealth	8	B
Race	1	B, E
Stroke	3	B
SkinCancer	3	B
Cancer	3	B
SleepTime	3	B
COPD	5	B
KidneyDisease	6	B
Diabetic	6	B
E-Cigarette	351208	D
MarijuanaUsage	271339	D
Weight	34948	C
Height	21734	C

Most features were categorical variables and were either ordinally encoded (e.g GeneralHealth) or nominally encoded (e.g. Race) by making use of One Hot Encoding due to machine learning algorithms assuming a relation between nearby values. A further feature reduction was performed by looking at a feature's correlation to heart disease with a cut-off set at 0.15, see Figure 2.1 and 2.2. This reduced the feature space to 5 significant features; HeartDisease, GeneralHealth, AgeGroup, Stroke, COPD, and DifficultyWalking.

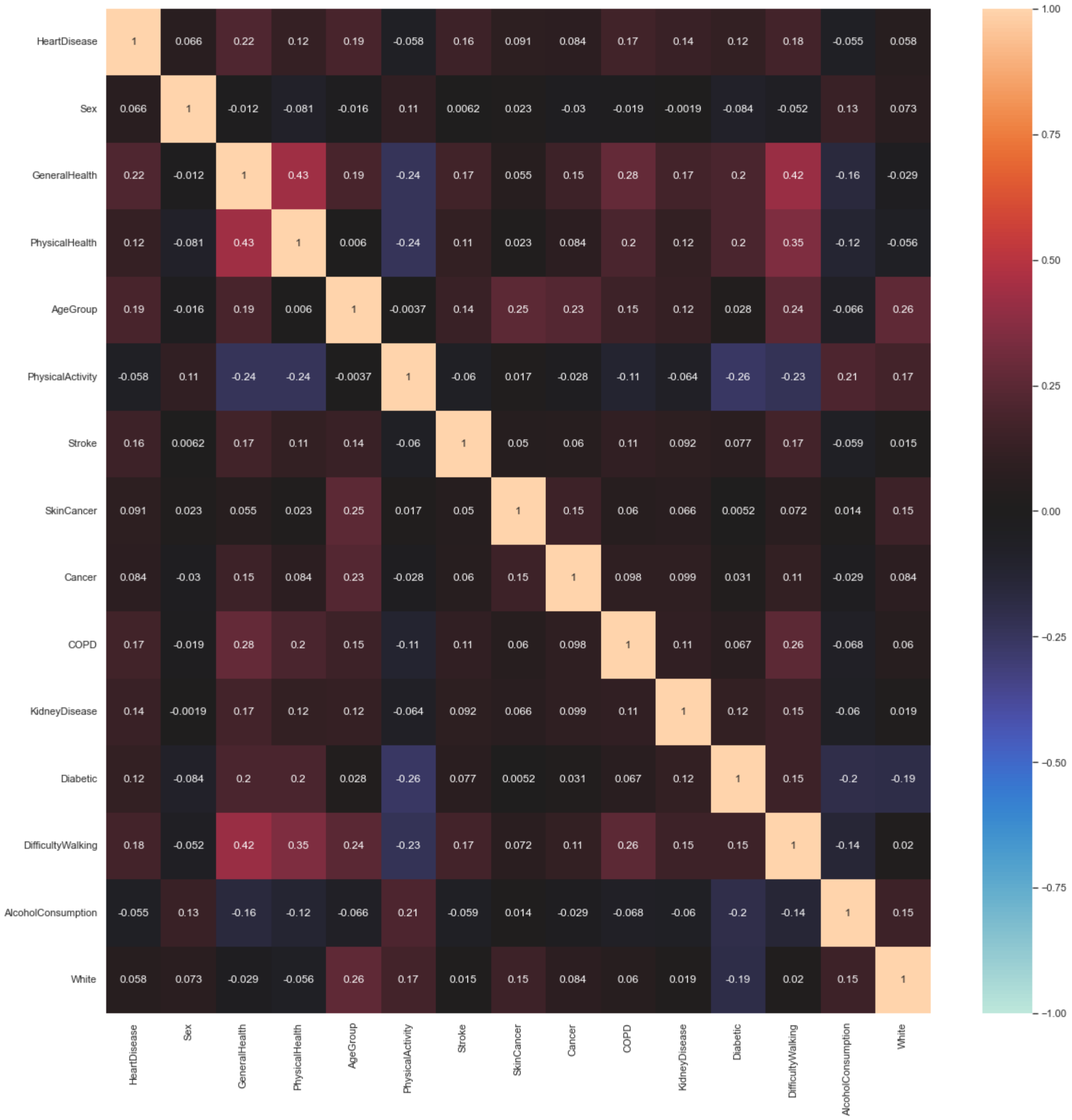

FIGURE 2.1: Feature Correlation Matrix

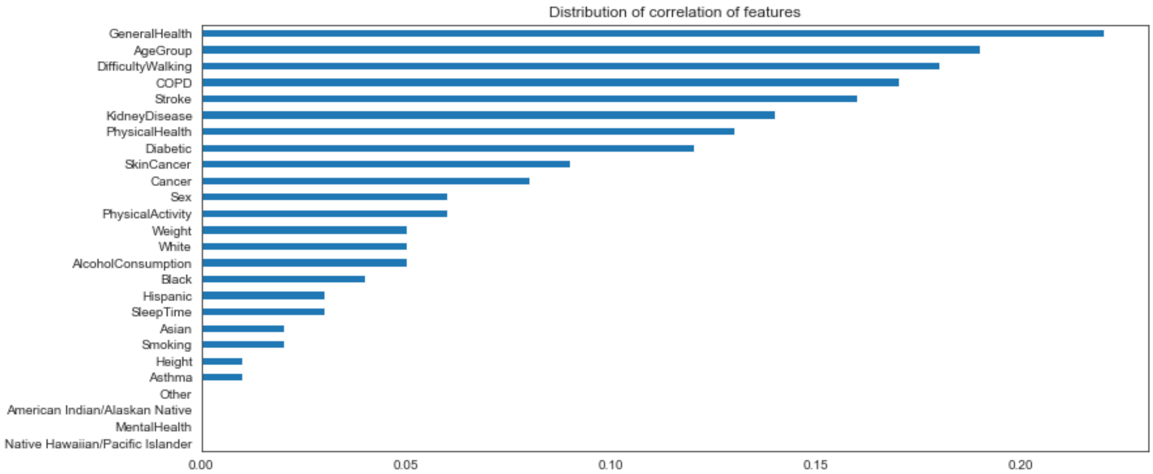

FIGURE 2.2: Distribution of Feature Correlations

2.3 Methods

The CDC survey data was in a SAS Transport (.xpt) file type format but due to the large file size (848 Mb), training and implementation of algorithms would have been impractically slow if the "xport" package was used. To remedy this, the data was converted and Pandas library was used in Jupyter Notebook instead.

Initially, the models performed very poorly due to machine learning algorithms being sensitive to the balance of training labels, undersampling was thus performed to improve class balance since only 5,6 percent of entries had heart disease. Scikit learn and Imblearn packages were used to implement undersampling. Samples were shuffled and split with the ratio 80-20 for training and testing respectively with the training set being further split into a validation set.

Sampling ratio used:

$$\alpha_u s = N_m / N_r M \tag{2.1}$$

A variety of machine learning algorithms were implemented and compared to find the best model for predicting heart disease. Principal Component Analyses (PCA), Logistic Regression, K-Nearest Neighbours (KNN), Random Forest, Naive Bayes Classifier, Multi-layer Perceptron Classifier, and Easy Ensemble Classifier algorithms

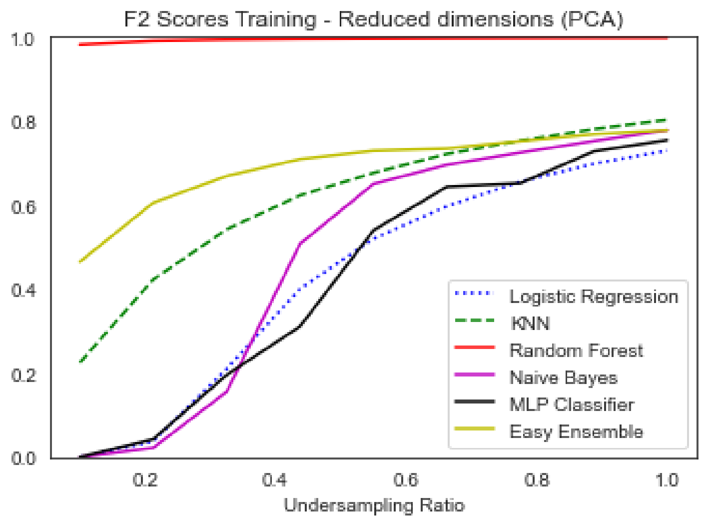

FIGURE 2.3: F2 Scores Training Set

were implemented and compared to find the best results. Due to the nature of the problem, Recall was prioritised over Precision and F2 Scores were thus used to evaluate the best fit model.

2.4 Analysis

F2 Scores for both the Training and Validation sets were compared and analysed for all the implemented algorithms after a PCA dimensionality reduction was done to identify the most promising model for further optimisation, see Figure 2.3 and 2.4. After implementation, training results performed worse than expected due to the overfitting of the data but Logistic Regression and MLP Classifier algorithms displayed the best performance.

The final Logistic Regression model very accurately predicted True Negatives but also often predicted False Positives, see Figure 2.5. This was, however, acceptable since the predictions could be used by medical professionals as an indicator that the patient exhibits many factors that contribute to the development of heart disease. While the patient doesn't have heart disease, they will have a high risk of developing it due to having many significant contributing factors, potentially functioning as an early warning indicator for medical professionals that the patient is likely to develop the condition.

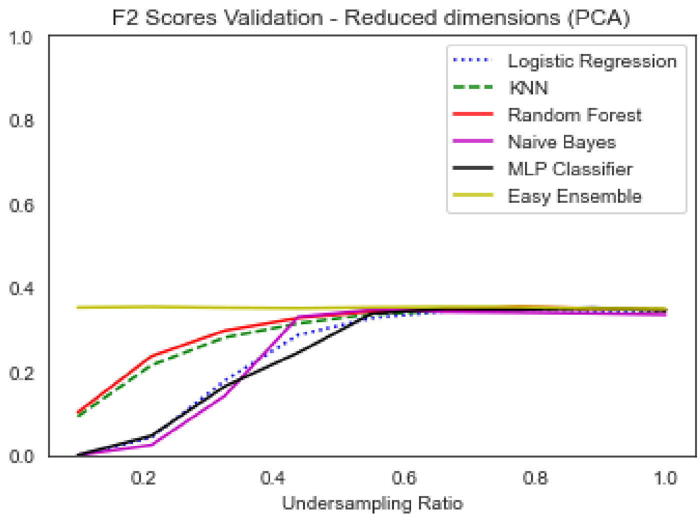

FIGURE 2.4: F2 Scores Validation Set

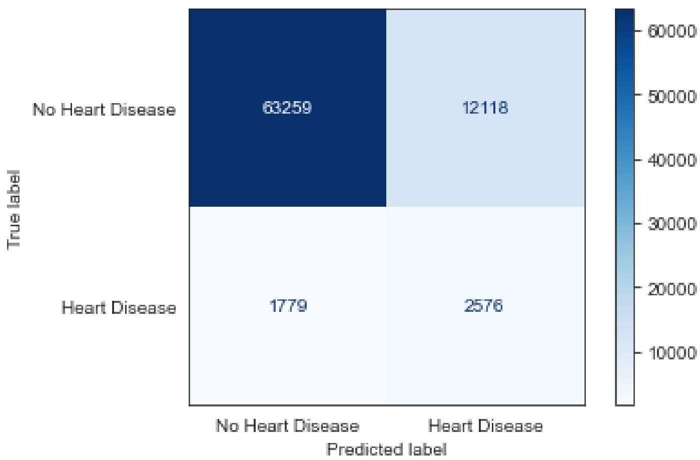

FIGURE 2.5: Logistic Regression Confusion Matrix

Chapter 3

Results and Discussion

3.1 Final Model

The final Logistic Regression model with initially reduced dimensions can be see in Equation 3.1 and Figure 3.1. An attempt was made to change the regularisation parameter using grid search but no improvements were detected. The model performed the best in terms of F2 Scores and was able to very accurately predict True Negatives but due to an imbalanced class problem, only 5,6 percent of sample entries had heart disease, the model could not accurately predict True Positives, see Figure 2.5. While the model often predicted False Positives, this was seen as beneficial since it could be used as an indicator to medical professionals that the patient exhibited many high-risk factors that contribute to developing heart disease and can thus be used as an early pre-emptive indicator that the patient has high risk of developing the condition.

$$ln\frac{P(HD)}{1-P(HD)} = 0.633GH + 0.271AG + 1.013S + 0.684C + 0.295DF - 5.012 \quad (3.1)$$

$HD = \text{HeartDisease}$
$GH = \text{GeneralHealth}$
$AG = \text{AgeGroup}$
$S = \text{Stroke}$
$C = \text{COPD}$
$DF = \text{DiffWalk}$

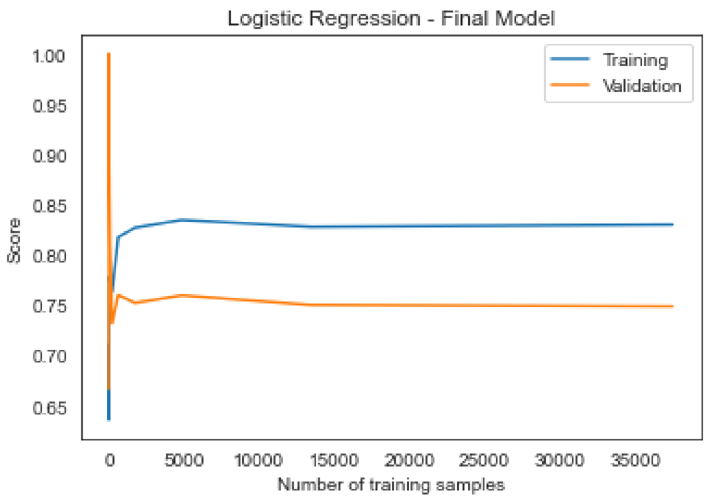

FIGURE 3.1: Final Model

3.1.1 Significant Features

During feature exploration and selection, significant features that contribute to heart disease were identified and analysed with a patient's general health being the primary indicator. Patients with poor general health had a significantly higher likelihood of developing heart disease with already present chronic obstructive pulmonary disease (COPD) also playing a significant role. The likelihood was further increased if the patient has experienced a stroke in the past and had difficulty walking or moving. Finally, a patient's age also plays a role in the development of heart disease, the older the patient is, the more likely they are to develop heart disease.

3.1.2 F2 Scores and Recall

The study focused on F2 Scores that are weighted to recall to determine the best model, see Equation 1.2. F2 Scores were prioritised over F1 Scores due to only 5,6 percent of sample entries having heart disease, it is thus more important to give greater weight to the model's recall than precision. The Logistic Regression algorithm displayed the best F2 scores compared to other algorithms, see Table 3.1.

TABLE 3.1: Logistic Regression F2 Scores

Model Set	F2 Score
Training	0.614
Validation	0.422
Test	0.401

3.2 Summary

Using 2022 survey data collected by the CDC, a machine-learning algorithm was constructed to predict heart disease in patients. After implementing dimensionality reduction using PCA on a variety of algorithms, the best fit model was selected based on F2 scores. A Logistic Regression algorithm was selected and was significantly better at predicting heart disease compared to a random sampler (59 percent vs. 50 percent). Algorithms were trained and optimised in the context of an unbalanced classification problem.

Chapter 4

Conclusions and Future Work

4.1 Conclusions

Machine learning can effectively be used to assist medical professionals in identifying significant risk factors that lead to the development of heart disease as well as predicting which patients have a high risk of developing the condition. A Logistic Regression algorithm with reduced dimensionality using PCA displayed the best results when looking at F2 scores, Recall, when making predictions about heart disease. A class imbalance problem could affect the ability of algorithms to accurately predict True Positives, additional survey data of patients with heart disease is thus required to improve the model's accuracy and optimisation.

4.2 Future Work

Potential future work could include making use of additional datasets from different countries to improve the generalisability of the model and for further model optimisation due to an increase in data points. According to Gola, et al. (2005), there exists a relationship between the use of growth hormones and cardiovascular health. The use of growth hormones could be a significant feature that could improve the model's predictions. Exploring and including additional features that could have an impact on the development of heart disease could further improve the model. Lastly, gathering more survey data from patients that suffer from heart disease would improve the class imbalance problem and potentially remove the need for under-sampling, further optimising the model.

Bibliography

Adams, A., Bojara, W., & Schunk, K. (2018). Early Diagnosis and Treatment of Coronary Heart Disease in Asymptomatic Subjects With Advanced Vascular Atherosclerosis of the Carotid Artery (Type III and IV b Findings Using Ultrasound) and Risk Factors. *Cardiology Research*, 22-27.

Brownlee, J. (2020). A Gentle Introduction to the Fbeta-Measure for Machine Learning. *Imbalanced Classification*.

Brownlee, J. (2020). Random Oversampling and Undersampling for Imbalanced Classification. *Imbalanced Classification*.

Centers for Disease Control and Prevention (CDC). (2022, November 29). *About Heart Disease*. Retrieved from Centers for Disease Control and Prevention: https://www.cdc.gov/heartdisease/about.htm

Duprez, D. (2006). Early detection of cardiovascular disease - the future of cardiology? *e-Journal of Cardiology Practice*.

Géron, A. (2019). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd Edition.* O'Reilly Media, Inc.

Gola, M., Bonadonna, S., Doga, M., & Giustina, A. (2005). Growth Hormone and Cardiovascular Risk Factors. *The Journal of Clinical Endocrinology & Metabolism 3*, 1864-1870.

Habehh, H., & Gohel, S. (2021). Machine Learning in Healthcare. *Current Genomics*, 291-300.

Jaadi, Z. (2022, September 26). *A Step-by-Step Explanation of Principal Component Analysis (PCA)*. Retrieved from Built In: https://builtin.com/data-science/step-step-explanation-principal-component-analysis

Javaid, M., Haleem, A., Singh, P. R., Suman, R., & Rab, S. (2022). Significance of machine learning in healthcare: Features, pillars and applications. *International Journal of Intelligent Networks*, 58-73.

Kumar, Y., Koul, A., Singla, R., & Ijaz, M. F. (2022). Artificial intelligence in disease diagnosis: a systematic literature review, synthesizing framework and future research agenda. *J Ambient Intell Humaniz Comput*, 1-28.

Selik, M. (2022, February 16). *xport 3.6.1*. Retrieved from The Python Package Index: https://pypi.org/project/xport/

Verma, Y. (2021, September 6). *A Complete Guide to Categorical Data Encoding*. Retrieved from Analytics India Mag: https://analyticsindiamag.com/a-complete-guide-to-categorical-data-encoding/

Wong, W. (2020, April 7). *Think twice before you use Principal Component Analysis in supervised learning tasks*. Retrieved from Towards Data Science: https://towardsdatascience.com/think-twice-before-you-use-principal-component-analysis-in-supervised-learning-tasks-70fbb68ebd0c